Alexander Taveira Blomenhofer

Curriculum Vitae

Education

2022 **Dr. rer. nat. in Mathematics**, *Universität Konstanz*, *summa cum laude*, Doctoral thesis:
"Gaussian mixture separation and denoising on parameterized varieties".
Advisors: Prof. Dr. Markus Schweighofer and Prof. Dr. Mateusz Michałek

2018 **M. Sc. Mathematics**, *Universität Konstanz*, 1.0.
Master's thesis: "A new algorithm for overcomplete tensor decomposition based on sums-of-squares optimisation" supervised by Prof. Dr. Markus Schweighofer.

2016 **B. Sc. Mathematics**, *Universität Konstanz*, 1.0.
Bachelor's thesis: "Erweiterte Formulierungen für Polygone" (Extended formulations for polygons, 2016) supervised by Prof. Dr. Markus Schweighofer.

2013 **Abitur (high school graduation)**, *Droste-Hülshoff-Gymnasium Meersburg*.

Employment

Sep 2026 - * **Assistant Professor of Geometry**, *Department of Mathematics*, University of Porto.

July 2024 - Aug 2026 **Postdoctoral Researcher at QMATH centre**, *supported by the ERC grant of Matthias Christandl*, University of Copenhagen.

Jan 2023 - June 2024 **Postdoctoral Researcher at CWI in Amsterdam**, *supported by the NWO grant OPTIMAL of Monique Laurent*.

2018-2022 **Research Assistant**, *Dep. Mathematics and Statistics, Universität Konstanz*.

Teaching

Winter 2026 **Statistical Inference**, *with Eliana Duarte*, University of Porto.

Winter 2026 **Linear algebra and Analytic Geometry**, *with ....*, University of Porto.

Winter 2025 **Commutative Algebra**, *with Nidhi Kaihnsa and Asbjørn Nordentoft*, University of Copenhagen.

Spring 2025 **Commutative Algebra**, *joint teaching with Nidhi Kaihnsa, exercises by Florian Riedel*, University of Copenhagen.

Spring 2024 **Mastermath course on Semidefinite Optimization**, *with Monique Laurent and Fernando de Oliveira Filho*, VU Amsterdam.

2022/23-WT **TA: Linear Algebra I**, Univ. Konstanz.

2022-ST **TA: Linear Algebra II**, Univ. Konstanz.

2021/22-WT **TA: Linear Algebra I**, Univ. Konstanz.

2021/22-WT **TA: Geometry of Linear Matrix Inequalities**, Univ. Konstanz.

2021-ST **TA: Real Algebraic Geometry II**, Univ. Konstanz.

2021-ST **Advanced Data and Information Literacy Track (ADILT) course "Vectors, Matrices and Tensors for Data Analysis with Julia"**, *with Markus Schweighofer*, Univ. Konstanz.

2020/21-WT **TA: Real Algebraic Geometry I**, Univ. Konstanz.

2020/21-WT **TA: Algorithmic Spectral Graph Theory**, Univ. Konstanz.

2020-ST **TA: Commutative Algebra**, Univ. Konstanz.

2019/20-WT **TA: Algorithmic Algebraic Geometry**, Univ. Konstanz.

2019-ST **TA: Polynomial Optimization**, Univ. Konstanz.

2019-ST **TA: Number Theory**, Univ. Konstanz.

2018/19-WT **TA: Introduction to Algebra**, Univ. Konstanz.

2015-2018 **Student teaching assistant for various courses, including Linear Algebra I-II, Algebra and Number theory**, Univ. Konstanz.

Organization

2023-2024 **Networks & Optimization Seminar**, biweekly seminar at CWI, with Sander Borst, Hilde Verbeek and Danish Kashaev, CWI, Amsterdam.

2019-2020 **Seminar on Real Geometry and Algebra**, *w. Markus Schweighofer*, Uni Konstanz.

Supervision

2018-2024 **Co-supervision**, *of various bachelor and masters students of Markus Schweighofer*, Universität Konstanz.

Publications

2025 **Moment-sos and spectral hierarchies for polynomial optimization on the sphere and quantum de Finetti theorems**, with Monique Laurent.
arXiv:2412.13191. *In press for SIAM J. Optim.*

2025 **On uniqueness of power sum decomposition**
SIAM J. Appl. Algebra Geom. 9(1), 211–234, DOI: 10.1137/23M1573355

2025 **Gaussian mixture identifiability from degree 6 moments**
Algebraic Statistics 16-1, 1–28. DOI: 10.2140/astat.2025.16.1

2023 **Identifiability for mixtures of centered Gaussians and sums of powers of quadratics**, with Alex Casarotti, Mateusz Michałek and Alessandro Oneto.
Bull. Lond. Math. Soc. 55, 5 (2023), 2407–2424, id: BLMS12871.

2022 **Ideals of spaces of degenerate matrices**, with Mateusz Michałek and Julian Vill.
Linear Algebra and its Applications 648 (2022), 56–69 (LAA-D-21-00505R1).

Preprints

2025 **A linear-time algorithm for Chow decompositions**, with Benjamin Lovitz. arXiv:2509.10450

2025 **On Defectivity of Joins, Reducible Secants and Fröberg's Conjecture**, with Alex Casarotti. arXiv:2509.10443

2023 **Nondefectivity of invariant secant varieties**, with Alex Casarotti. arXiv:2312.12335. *Under review for Selecta Math.*

Research stays

May 2026 **Research Stay**, *Max-Planck-Institut für Mathematik in den Naturwissenschaften*, Leipzig.
Invited by Bernd Sturmfels for two weeks to discuss tensor rank and secant varieties with members of the institute.

Aug 2025 **Research Stay**, *Università di Ferrara*.
Invited by Alex Casarotti for a week to work on a joint project about secant nondefectivity.

July 2025 **Research Stay**, *Università di Trento*.
Invited by Fulvio Gesmundo to work on the MultilinearVerse project.

June 2025 **Research Visit**, *INRIA Cotê d'Azur*, Nice.
Invited by Khazhgali Kozhasov to discuss ideas on polynomial optimisation.

Oct 2024 **Research Stay**, *CWI*, Amsterdam.
I was invited by Monique Laurent at CWI for three weeks to work with her and Matthias Christandl on Quantum de Finetti theorems and polynomial optimization hierarchies.

Nov 2023 **Research Stay**, *Università di Trento*.
I was invited by Alessandro Oneto at the University of Trento for a week to present and discuss a partial solution of the Baur-Draisma-de Graaf conjecture.

Talks, conferences and other venues

June 2026 **SIAM Conference on Optimization**, Edinburgh.
Invited minisymposium talk: *Fast tensor decomposition via contraction varieties.*

May 2026 **Positive geometry seminar**, Max-Planck-Institut für Mathematik in den Naturwissenschaften, Leipzig.
Seminar talk: *Border rank = rank for Kruskal tensors.*

April 2026 **Algebra, Combinatorics and Number theory seminar**, University of Porto.
Seminar talk: *Border rank = rank for Kruskal tensors.*

Nov 2025 **Geometry of tensors mini-workshop**, *Institut Élie Cartan de Lorraine*, Nancy.
Invited talk: *Additive X-rank decomposition.*

Sep 2025 **TENORS Workshop**, *Universität Konstanz*, Konstanz.
Invited talk: *Recent advances in tensor decomposition and identifiability.*

Aug 2025 **Computations in Algebraic Geometry**, *ETH Zürich*, Zürich.
Poster: *The contraction variety of a tensor.*

July 2025 **SIAM AG**, Madison, Wisconsin.
Invited minisymposium talk: *Moment-sos and spectral hierarchies for polynomial optimization on the sphere and quantum de Finetti theorems.*

July 2025 **INABAG conference**, *Università di Trento*, Trento.

July 2024 **MEGA 2024**, *MPI-MiS*, Leipzig.
Poster: *Dimensions of invariant secant varieties.*

April 2024 **Dutch Mathematical Congress**, *TU Delft*.
Invited talk for AIM: *Algebraic Geometry in Machine Learning*.

March 2024 **MoPAT24**, *Universität Konstanz*, Conference on polynomial and tensor optimization.
Invited talk: Nondefectivity of invariant secant varieties

Oct 2023 **Chow Lectures**, *MPI-MiS*, Leipzig.

Sep 2023 **Conference on Applied Algebra**, *Universität Osnabrück*.
Poster: *Identifiability of Gaussian mixtures from sixth-order moments*.

July 2023 **SIAM AG**, *TU Eindhoven*, Eindhoven.
Invited minisymposium talk: *Unique powers-of-forms decompositions from simple Gram spectrahedra*.

June 2023 **FOCM**, *Sorbonne Université*, Paris.
Poster in the Data Science and Machine Learning session (II.4):
Identifiability of Gaussian mixtures from sixth-order moments.

May 2023 **OPTIMAL meeting**, *TU Delft*, Delft.
Talk: *Unique power sum decompositions from simple Gram spectrahedra*.

April 2023 **DIAMANT symposium**, *De Zalen van Zeven*, Utrecht.
Talk: *Identifiability of Gaussian Mixtures from their sixth moments*

April 2023 **Dutch Mathematical Congress**, *Van der Walk Hotel*, Utrecht.

Mar 2023 **Real Algebraic Geometry with a View toward Koopman Operator Methods**, *Mathematisches Forschungsinstitut Oberwolfach*, Oberwolfach.
Talk: *A semidefinite algorithm for powers-of-forms decomposition*.

2022 **Workshop on Semidefinite and Polynomial Optimization**, *CWI*, Amsterdam.

2022 **International Conference on Continuous Optimization (ICCOPT)**, *Lehigh University*, Bethlehem, PA, USA.
Invited minisymposium talk: *Projecting towards the image set of a polynomial map with Sum-of-Squares Relaxations*.

2022 **MEGA 2022**, *Pedagogical University of Kraków*, Kraków, Poland.
Talk: *Identifiability for Mixtures of Gaussians* (joint work with A. Casarotti, M. Michałek and A. Oneto).

2020 **Real Algebraic Geometry with a View Toward Hyperbolic Programming and Free Probability**, *Mathematisches Forschungsinsitut Oberwolfach*, Oberwolfach.

2019 **SIAM Conference on Applied Algebraic Geometry**, *Universität Bern*, Bern.

2019 **Arctic Applied Algebra**, *University of Tromsø*, Tromsø.
Talk on the topic of my master's thesis.

2018 **International Conference on Polynomial and Tensor Optimization**, *Xiangtan University*, Xiangtan (Hunan Province, PR China).
Talk on the topic of my master's thesis.

2018 **Summer School on Numerical Computing in Algebraic Geometry**, *Max Planck Institute for Mathematics in the Sciences*, Leipzig.

Services

2019-* **Reviewer for zbMath**.

2024-* **Peer reviewing for SIAM J. Appl. Algebra Geom., Math. Ann., QIP and other journals and conferences**.

Programming Skills

Routined JULIA, particularly `JuMP`, `MultivariatePolynomials`, `SumOfSquares` and generally multilinear algebra and sum-of-squares optimization. Also `DataFrames`, `LIBSVM` etc. and partial experience with symbolic computation (`OSCAR`, `NEMO`...)

Intermediate PYTHON, JavaScript, LaTeX, Bash/Shell, Jupyter

Basic C, Java

Languages

Native **German**

Fluent **English, Portuguese, Dutch**

Basic **French, Danish**

Research Interests

- Secant Varieties
- Computational Complexity
- Machine Learning
- Computational Algebraic Geometry
- Invariant Theory
- Algebraic Statistics
- Quantum Information Theory
- Semidefinite Programming